Brief A: Early-Warning Model for Pump Health

Audience: Maintenance Engineering | Prepared by: Paul

Sensor logic

Vibration is our strongest signal. Healthy pumps average 1.9g. Critical pumps average 4.2g. A t-test confirms this gap is real, not noise ($p < 0.001$), and the two groups don't overlap at all.

Features used

- Rolling mean (36 hr) - smooths noisy readings, shows the trend.
- RMS of vibration - flags sudden spikes, not just the average.
- Peak-to-peak range - captures instability, not just level.
- Rolling std of pressure - often unstable before vibration is.

Model and results

Logistic regression on those four features plus temperature and motor current, predicting Healthy vs. Failing. Test set results:

- Accuracy: 95.5%
- Sensitivity (recall on failing pumps): 82.9% - 116 of 140 caught
- False positive rate: 1.1% - 6 of 520 healthy readings wrongly flagged
- False negatives: 24 failing pumps missed

Few false alarms, so techs won't chase phantom problems. But 1 in 6 failing pumps still slips through, so this should support routine inspection, not replace it, until recall improves.

Brief B: The Business Case for Early Pump Warnings

Audience: Finance / Executive | Prepared by: Paul

The problem

An unplanned pump failure costs us twice - once in emergency repairs, once in lost production. In a four-week sample, our 24 pumps showed 61 separate warning episodes. Most go uncaught until the pump has already failed.

What we built

A system that watches each pump's vibration, pressure, and temperature, and raises a flag when the pattern looks like past failures - like a car's dashboard warning light coming on before the engine actually stops.

Estimated value

It catches about 83 of every 100 pumps heading toward failure, with few false alarms. Using a typical cost of $15,000 per unplanned failure:

- Catching 10 failures early saves roughly $150,000 in repairs and downtime.
- Fewer surprise breakdowns means less emergency spending and less risk of bigger failures.
- False alarms are rare, so maintenance crews aren't sent out unnecessarily.

Bottom line: avoiding just a handful of failures a year pays for this system many times over. About 1 in 6 failing pumps still isn't caught yet, so treat this as an early pilot that improves over time - not a full replacement for inspections just yet.

Reflection

Switching between these two briefs was harder than expected - not the content, but what to leave out. The engineer's version could lean on the confusion matrix and let the numbers speak. The CFO version needed the same result - 24 missed pumps - turned into a dollar figure instead, with no "recall" or "logistic regression" at all. The story stayed the same underneath: catch problems early, save money, don't cry wolf too often. Trusting that the CFO didn't need the method to trust the conclusion took a couple of rewrites to get right.